
Supplementary Data for Layered representation diagnostics for ultra-short metagenomic reads

Ruixiang Mei¹ and Jianhua Huang¹

¹School of Data Science, The Chinese University of Hong Kong, Shenzhen, Shenzhen 518172, Guangdong, China

Abstract

Supplementary figures and source-table preview supporting the representation-diagnostic analyses.

Keywords supplementary data, metagenomics, representation diagnostics

Supplementary data overview

This supplementary file collects the visual audits and source-table preview that support the main representation-diagnostic manuscript. Figures S1–S3 evaluate baseline controls, empirical information proxies and sequencing-error-aware perturbations. Figures S4–S7 examine local mutation fraction, P-channel counterfactuals, MSP bin and weight sensitivity, kNN mutual-information robustness and dimension-matched high-k compressed baselines. Figures S8–S10 summarize P/MSP redundancy, runtime, relation audits and the CAMI II marine anonymous-read stability probe. Table S1 gives a compact preview of the CAMI II marine source table; the full CSV is retained in the accompanying data package.

Supplementary Figures

Supplementary Table

Supplementary Figure S1 | Baseline and mixed-metric audit

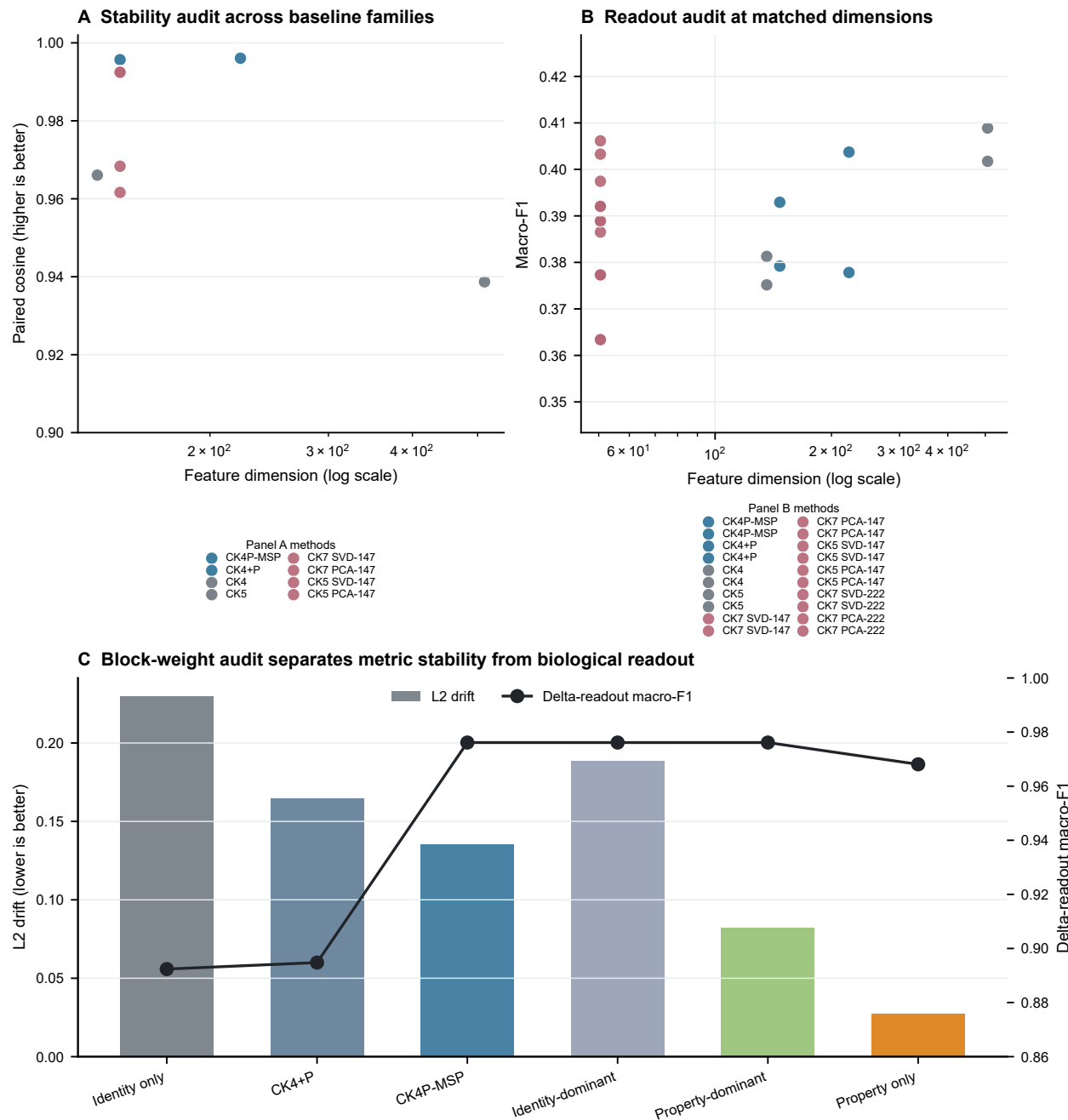


Figure S1 Baseline and mixed-metric audit.
Alt text: Baseline audit plots comparing full-position matrices, reduced controls and block-weight sensitivity.

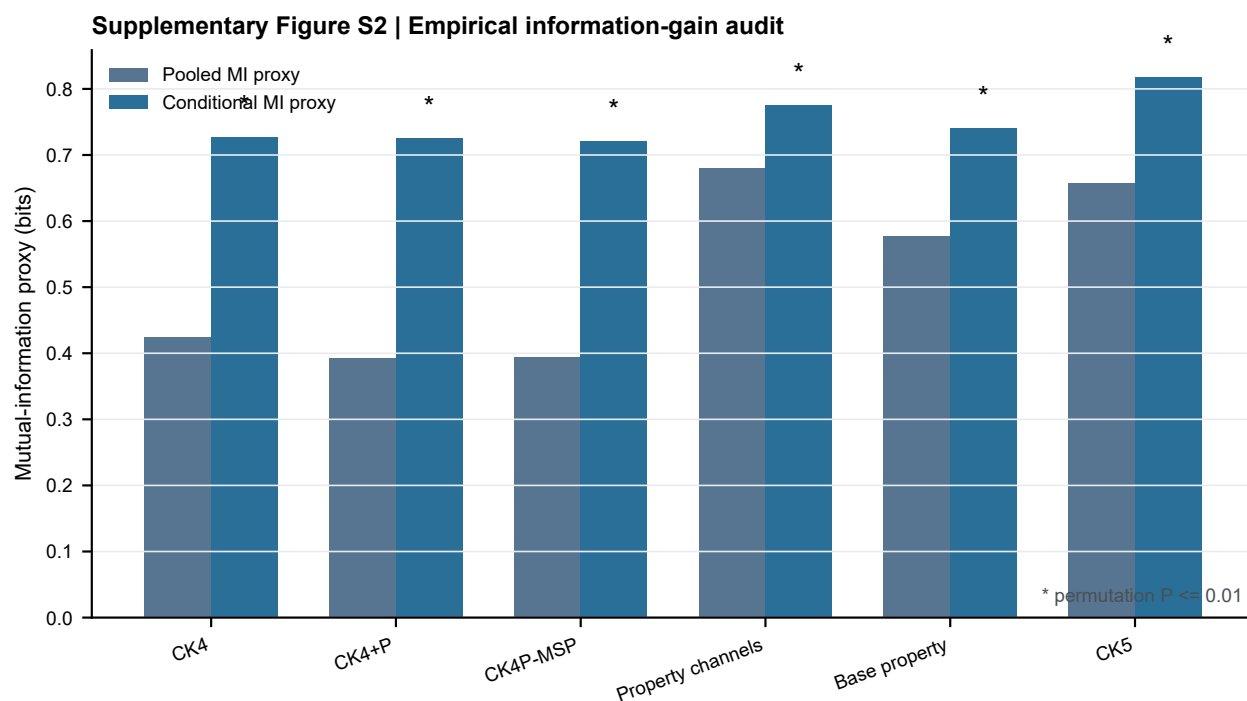


Figure S2 Empirical MI and conditional-MI audit.

Alt text: Bar plots summarizing empirical mutual-information and conditional-mutual-information audit values across representation blocks.

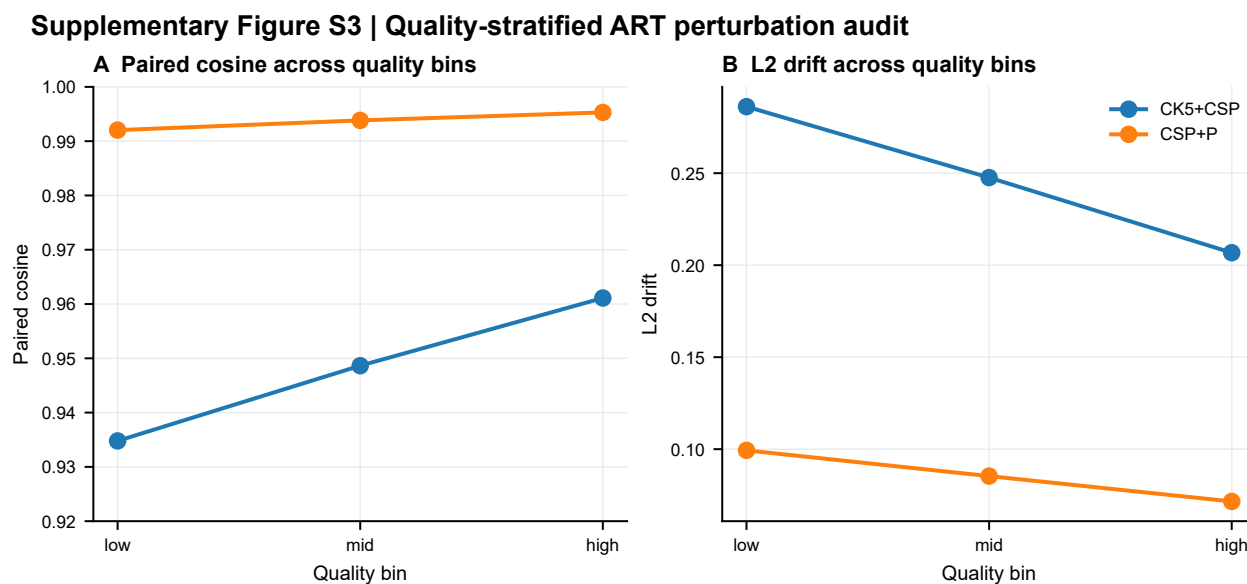


Figure S3 Quality-stratified ART perturbation audit.

Alt text: Line plots showing ART perturbation behavior across quality levels for paired cosine and L2 drift.

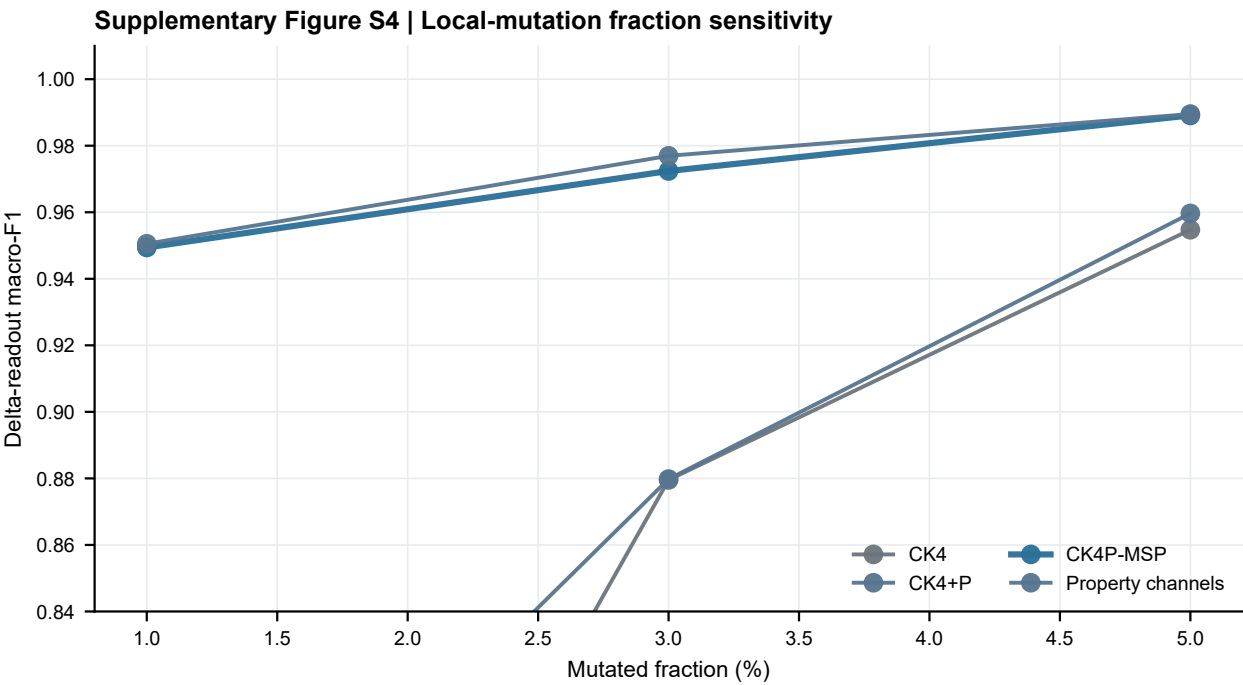


Figure S4 Local mutation-fraction sweep.
Alt text: Line plot showing local mutation-fraction sensitivity across compact representation families.

Table S1. Preview of the CAMI II marine subset source table. The full CSV is retained in the data package.

length	condition	method	features	paired cosine	L2 drift	top-1
69	N_3pct	CK4	136	0.9643	0.2651	1.0000
69	N_3pct	CK4+P	147	0.9944	0.1059	1.0000
69	N_3pct	CK4P-MSP	222	0.9948	0.1016	1.0000
69	N_3pct	CK4P-MSP+SD	297	0.9943	0.1061	1.0000
69	N_3pct	CK5	512	0.9382	0.3491	1.0000
69	N_3pct	CK5+CK4P-MSP	734	0.9665	0.2572	1.0000
69	N_3pct	CSP	147	0.9941	0.1080	1.0000
75	N_3pct	CK4	136	0.9683	0.2497	1.0000
75	N_3pct	CK4+P	147	0.9950	0.0992	1.0000
75	N_3pct	CK4P-MSP	222	0.9954	0.0951	1.0000
75	N_3pct	CK4P-MSP+SD	297	0.9950	0.0994	1.0000
75	N_3pct	CK5	512	0.9441	0.3322	1.0000
75	N_3pct	CK5+CK4P-MSP	734	0.9698	0.2444	1.0000
75	N_3pct	CSP	147	0.9948	0.1011	1.0000
100	N_3pct	CK4	136	0.9687	0.2483	1.0000

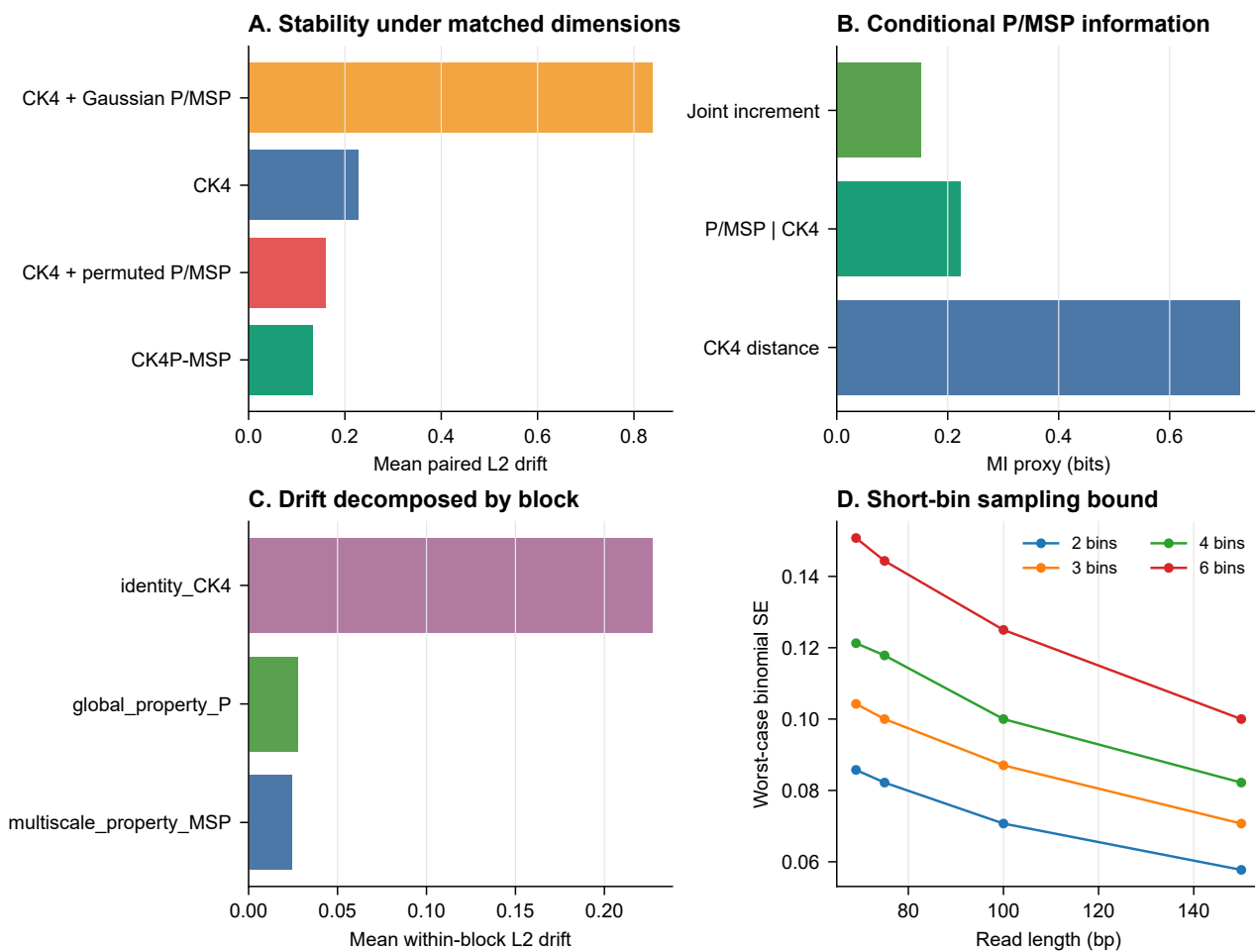


Figure S5 P-channel counterfactual and short-bin reliability audit.

Alt text: Multi-panel audit of P-channel counterfactual behavior, drift decomposition and short-bin sampling bounds.

Supplementary Figure S6 | MSP binset and gamma-sensitivity audit

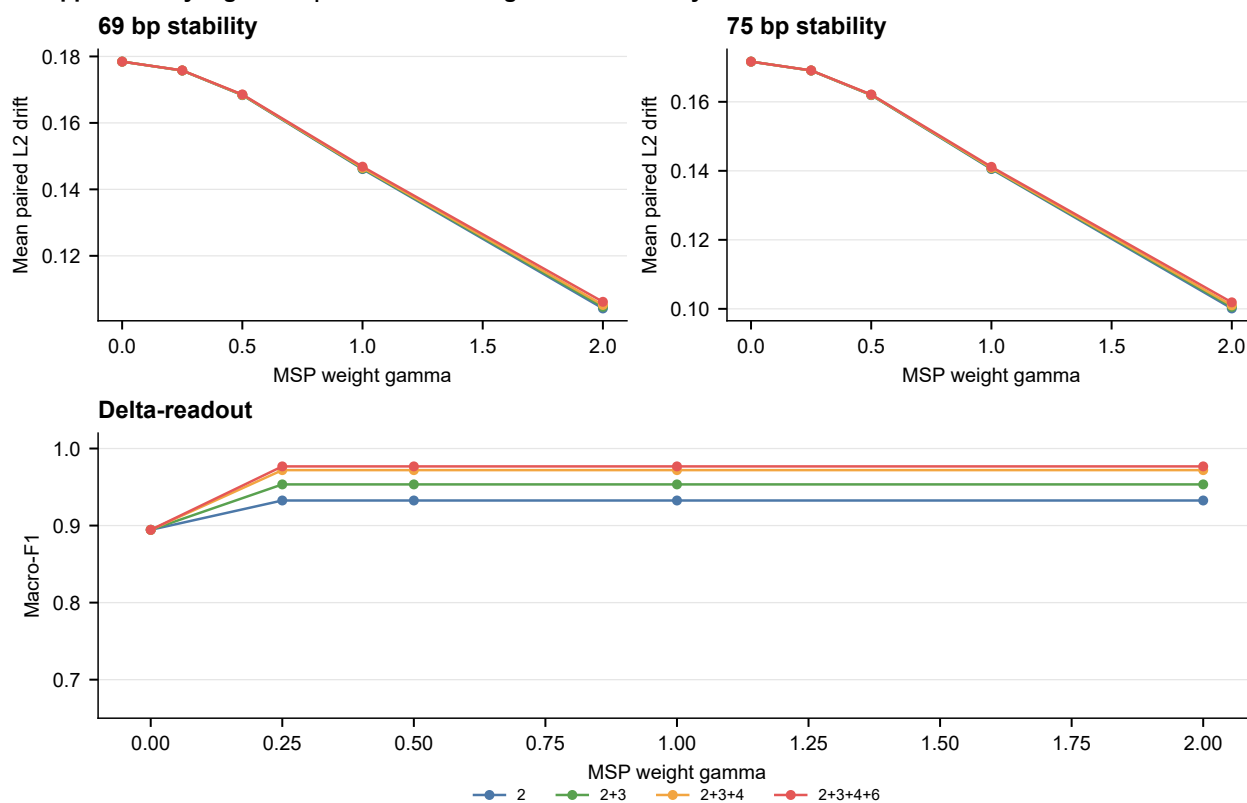
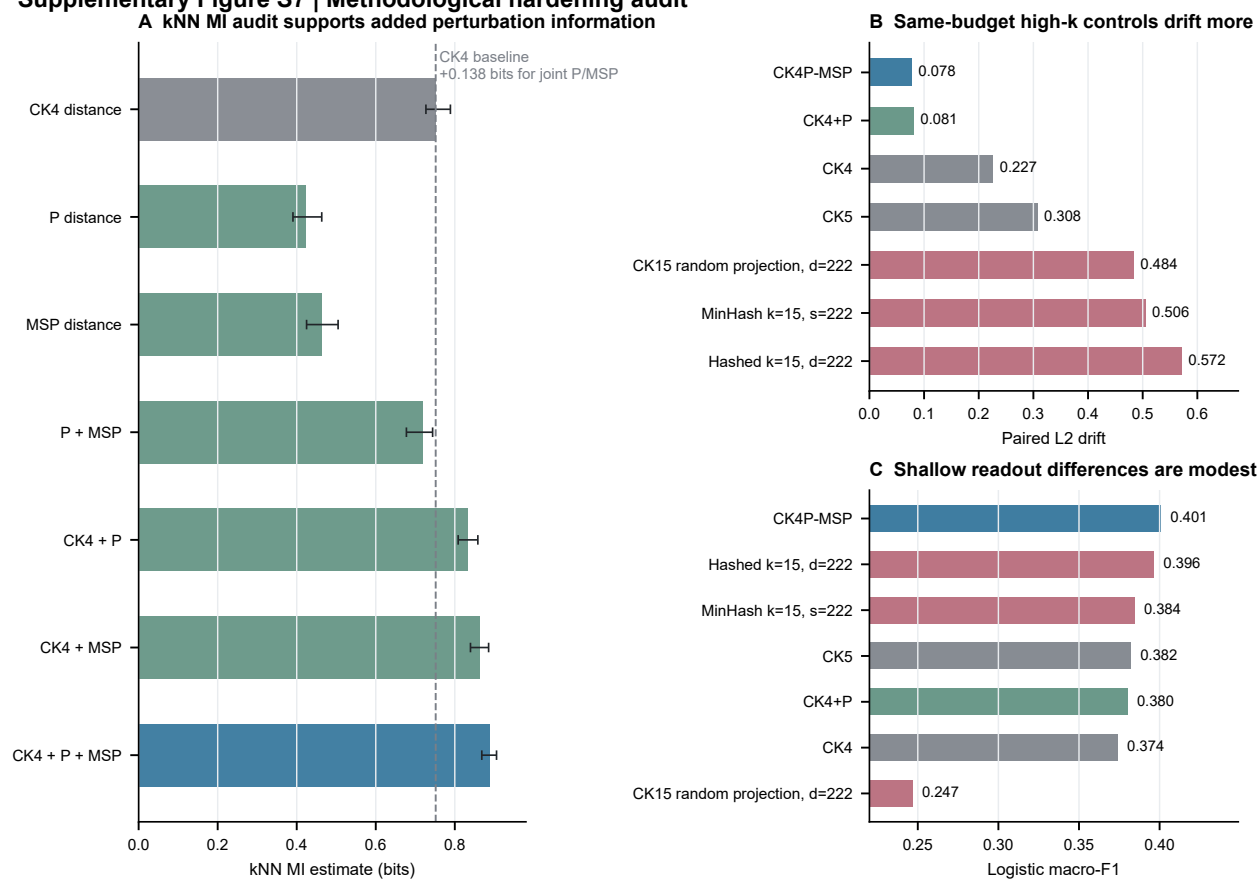


Figure S6 MSP binset and gamma-sensitivity audit.

Alt text: Line plots showing MSP binset and gamma-weight sensitivity for 69 and 75 bp reads and delta-readout.

Supplementary Figure S7 | Methodological hardening audit**Figure S7** kNN MI robustness and dimension-matched high-k compressed baseline audit.

Alt text: Multi-panel audit showing kNN mutual-information robustness and dimension-matched high-k compressed baseline comparisons.

Supplementary Figure S8 | P/MSP contribution, redundancy and runtime audit

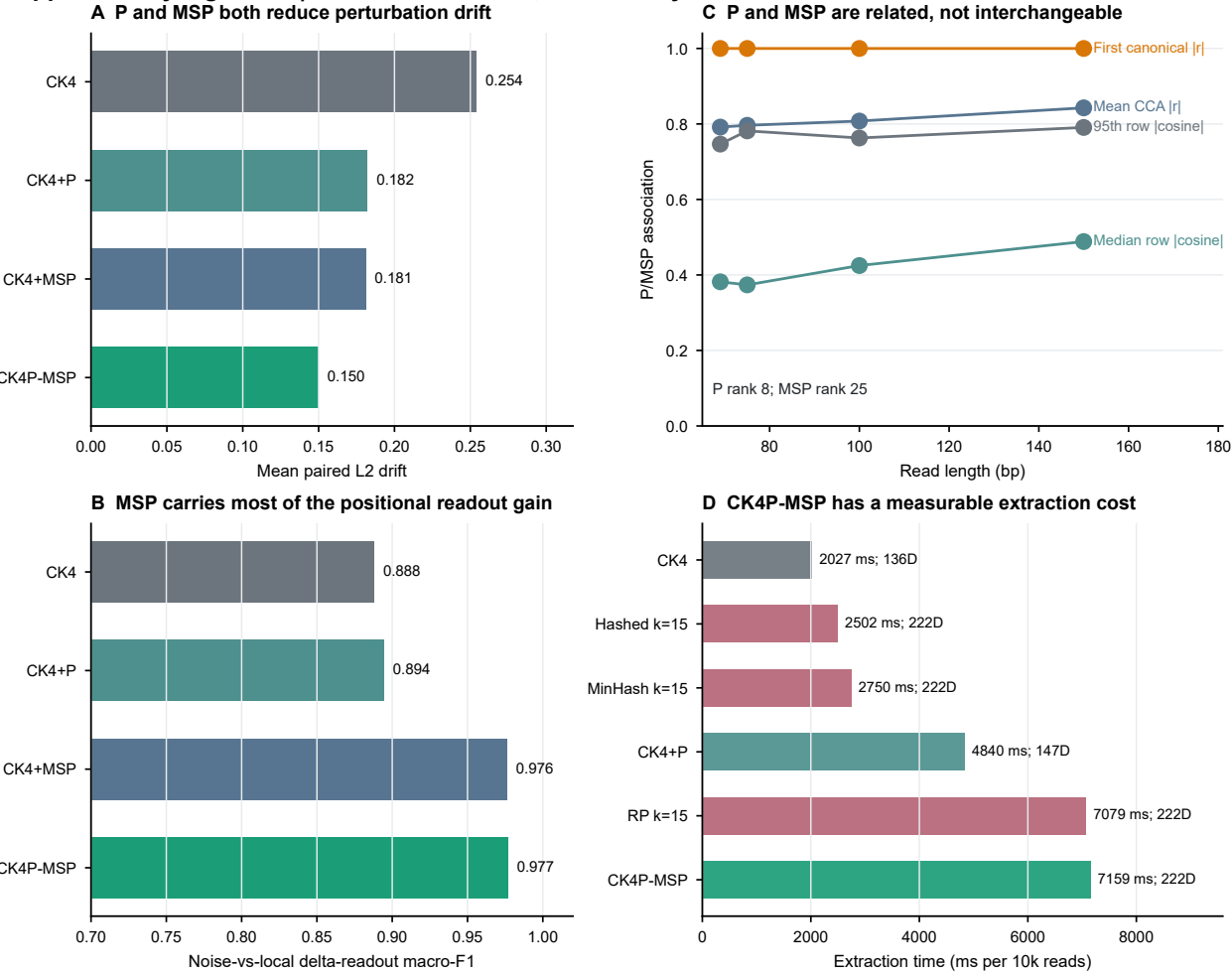


Figure S8 P/MSP contribution, redundancy and runtime audit.
Alt text: Multi-panel audit showing P and MSP contributions, redundancy and runtime costs.

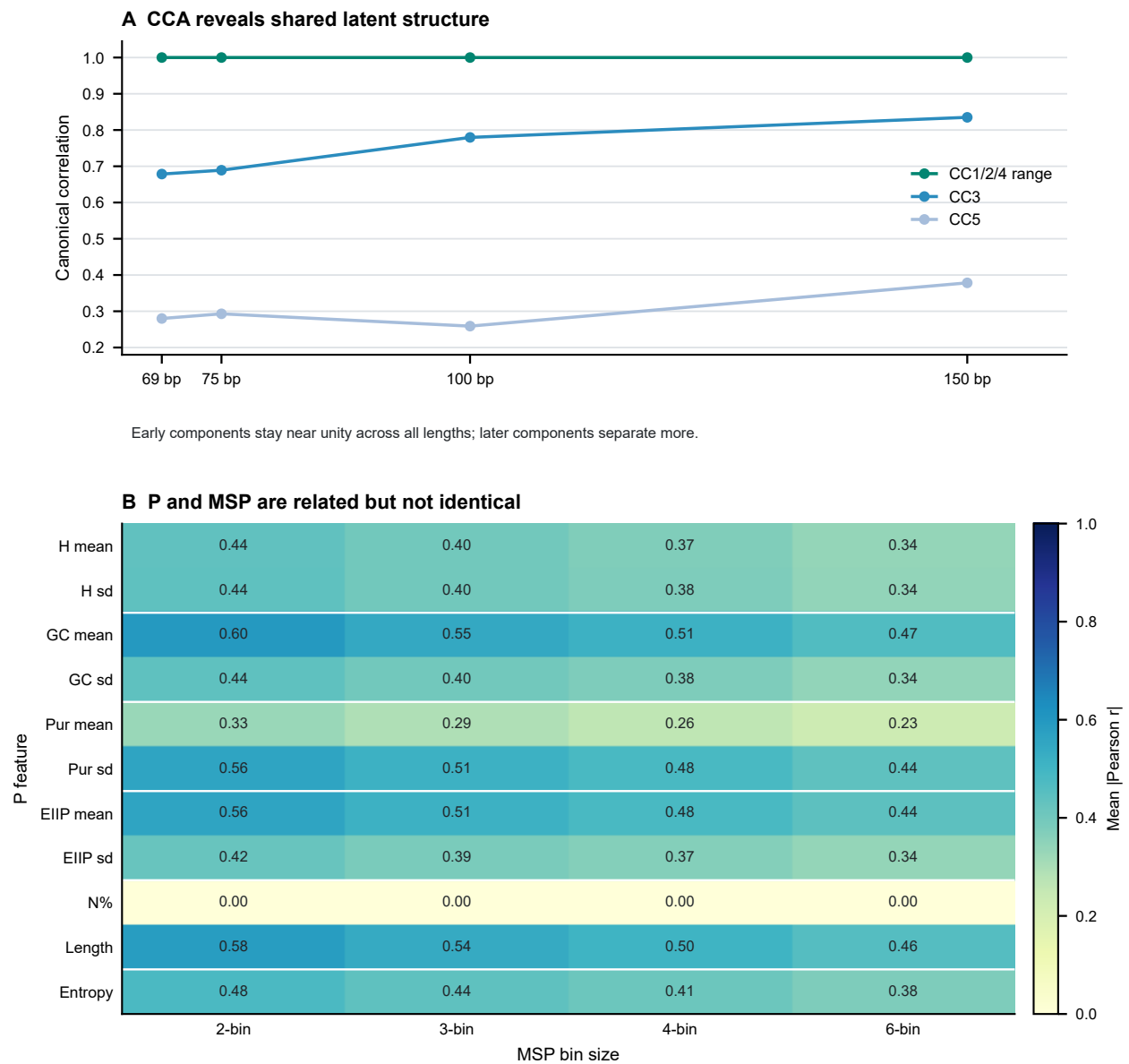


Figure S9 P/MSP relation audit.

Alt text: Line and heatmap panels showing CCA and correlation relationships between P and MSP feature layers.

Supplementary Figure S10 | CAMI II marine subset probe
n=4,000 reads; 48,000 length/condition rows; unlabeled perturbation-stability probe

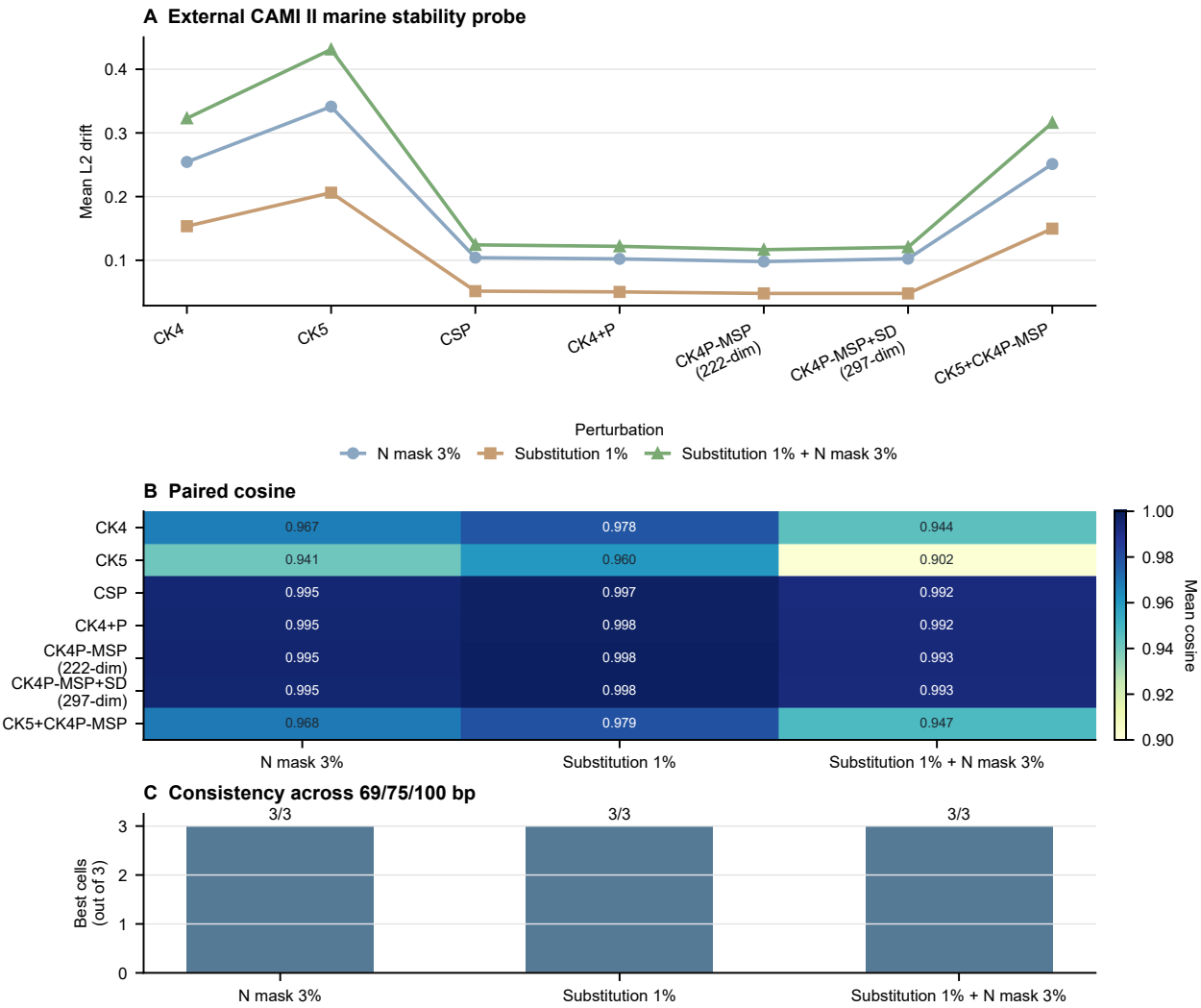


Figure S10 CAMI II marine anonymous-read stability probe.
Alt text: Line, heatmap and bar panels showing CAMI II marine anonymous-read stability across length and perturbation settings.